MOSFETs Silicon N-channel MOS (U-MOSⅧ-H)

TPH4R008NH

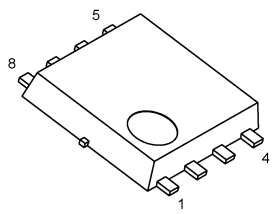
1. Applications

- DC-DC Converters
- Switching Voltage Regulators
- Motor Drivers

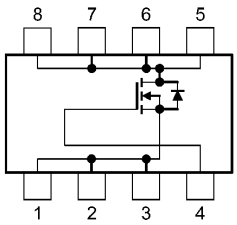
2. Features

(1) Small, thin package
(2) High-speed switching
(3) Small gate charge: Q_{SW} = 18 nC (typ.)
(4) Low drain-source on-resistance: $R_{DS(ON)}$ = 3.3 mΩ (typ.) (V_{GS} = 10 V)
(5) Low leakage current: I_{DSS} = 10 μA (max) (V_{DS} = 80 V)
(6) Enhancement mode: V_{th} = 2.0 to 4.0 V (V_{DS} = 10 V, I_D = 1.0 mA)

3. Packaging and Internal Circuit



SOP Advance



1, 2, 3: Source
4: Gate
5, 6, 7, 8: Drain

4. Absolute Maximum Ratings (Note) (T_a = 25°C unless otherwise specified)

Characteristics			Symbol	Rating	Unit
Drain-source voltage			V_{DSS}	80	V
Gate-source voltage			V_{GSS}	±20	
Drain current (DC)	(Silicon limit)	(Note 1), (Note 2)	I_D	100	A
Drain current (DC)	(T_c = 25°C)	(Note 1)	I_D	60	
Drain current (pulsed)	(t = 1 ms)	(Note 1)	I_{DP}	200	
Power dissipation	(T_c = 25°C)		P_D	78	W
Power dissipation	(t = 10 s)	(Note 3)	P_D	2.8	W
Power dissipation	(t = 10 s)	(Note 4)	P_D	1.6	W
Single-pulse avalanche energy		(Note 5)	E_{AS}	340	mJ
Avalanche current			I_{AR}	60	A
Channel temperature			T_{ch}	150	°C
Storage temperature			T_{stg}	-55 to 150	

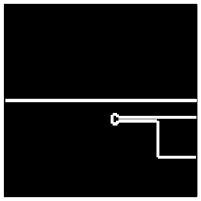
Note: Using continuously under heavy loads (e.g. the application of high temperature/current/voltage and the significant change in temperature, etc.) may cause this product to decrease in the reliability significantly even if the operating conditions (i.e. operating temperature/current/voltage, etc.) are within the absolute maximum ratings.
Please design the appropriate reliability upon reviewing the Toshiba Semiconductor Reliability Handbook ("Handling Precautions"/"Derating Concept and Methods") and individual reliability data (i.e. reliability test report and estimated failure rate, etc).

Start of commercial production
2013-02

5. Thermal Characteristics

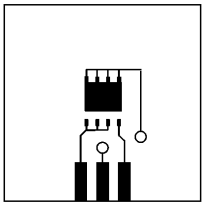
Characteristics			Symbol	Max	Unit
Channel-to-case thermal resistance	(T_c = 25°C)		$R_{th(ch-c)}$	1.60	°C/W
Channel-to-ambient thermal resistance	(t = 10 s)	(Note 3)	$R_{th(ch-a)}$	44.6	°C/W
Channel-to-ambient thermal resistance	(t = 10 s)	(Note 4)	$R_{th(ch-a)}$	78.1	°C/W

Note 1: Ensure that the channel temperature does not exceed 150°C.
Note 2: Limited by silicon chip capability. Package limit is 60 A.
Note 3: Device mounted on a glass-epoxy board (a), Figure 5.1
Note 4: Device mounted on a glass-epoxy board (b), Figure 5.2
Note 5: V_{DD} = 60 V, T_{ch} = 25°C (initial), L = 0.08 mH, R_G = 1.0 Ω, I_{AR} = 60 A



FR-4
25.4 × 25.4 × 0.8
(Unit: mm)

Fig. 5.1 Device Mounted on a Glass-Epoxy Board (a)



FR-4
25.4 × 25.4 × 0.8
(Unit: mm)

Fig. 5.2 Device Mounted on a Glass-Epoxy Board (b)

Note: This transistor is sensitive to electrostatic discharge and should be handled with care.

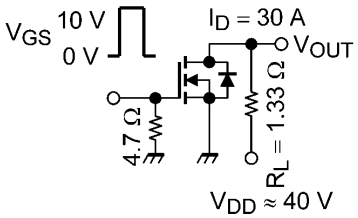
6. Electrical Characteristics

6.1. Static Characteristics (T_a = 25°C unless otherwise specified)

Characteristics	Symbol	Test Condition	Min	Typ.	Max	Unit
Gate leakage current	I_{GSS}	V_{GS} = ±20 V, V_{DS} = 0 V	—	—	±0.1	μA
Drain cut-off current	I_{DSS}	V_{DS} = 80 V, V_{GS} = 0 V	—	—	10	
Drain-source breakdown voltage	$V_{(BR)DSS}$	I_D = 10 mA, V_{GS} = 0 V	80	—	—	V
	$V_{(BR)DSX}$	I_D = 10 mA, V_{GS} = -20 V	60	—	—	
Gate threshold voltage	V_{th}	V_{DS} = 10 V, I_D = 1.0 mA	2.0	—	4.0	
Drain-source on-resistance	$R_{DS(ON)}$	V_{GS} = 10 V, I_D = 30 A	—	3.3	4.0	mΩ

6.2. Dynamic Characteristics (T_a = 25°C unless otherwise specified)

Characteristics	Symbol	Test Condition	Min	Typ.	Max	Unit
Input capacitance	C_{iss}	V_{DS} = 40 V, V_{GS} = 0 V, f = 1 MHz	—	4100	5300	pF
Reverse transfer capacitance	C_{rss}		—	32	64	
Output capacitance	C_{oss}		—	890	—	
Gate resistance	r_g	—	—	1.2	1.8	Ω
Switching time (rise time)	t_r	See Figure 6.2.1.	—	8.6	—	ns
Switching time (turn-on time)	t_{on}		—	25	—	
Switching time (fall time)	t_f		—	12	—	
Switching time (turn-off time)	t_{off}		—	52	—	



Duty ≤ 1%, t_w = 10 μs

Fig. 6.2.1 Switching Time Test Circuit

6.3. Gate Charge Characteristics (T_a = 25°C unless otherwise specified)

Characteristics	Symbol	Test Condition	Min	Typ.	Max	Unit
Total gate charge (gate-source plus gate-drain)	Q_g	V_{DD} ≈ 40 V, V_{GS} = 10 V, I_D = 60 A	—	59	—	nC
Gate-source charge 1	Q_{gs1}		—	18	—	
Gate-drain charge	Q_{gd}		—	12	—	
Gate switch charge	Q_{SW}		—	18	—	

6.4. Source-Drain Characteristics (T_a = 25°C unless otherwise specified)

Characteristics	Symbol	Test Condition	Min	Typ.	Max	Unit
Reverse drain current (pulsed) (Note 6)	I_{DRP}	—	—	—	200	A
Diode forward voltage	V_{DSF}	I_{DR} = 60 A, V_{GS} = 0 V	—	—	-1.2	V

Note 6: Ensure that the channel temperature does not exceed 150°C.

7. Marking

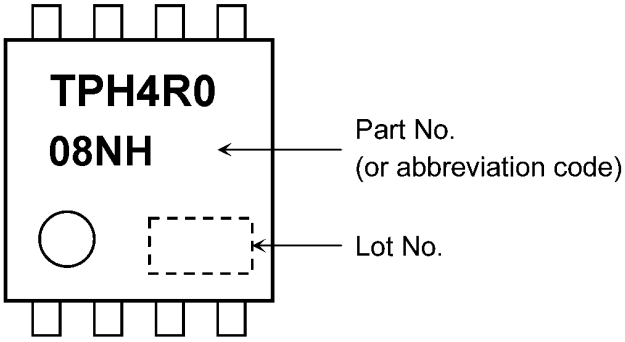


Fig. 7.1 Marking

8. Characteristics Curves (Note)

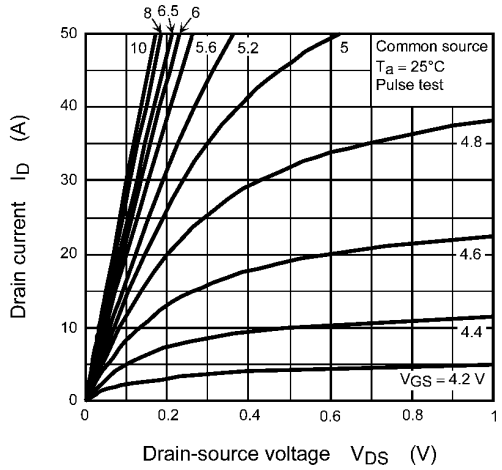


Fig. 8.1 I_D - V_{DS}

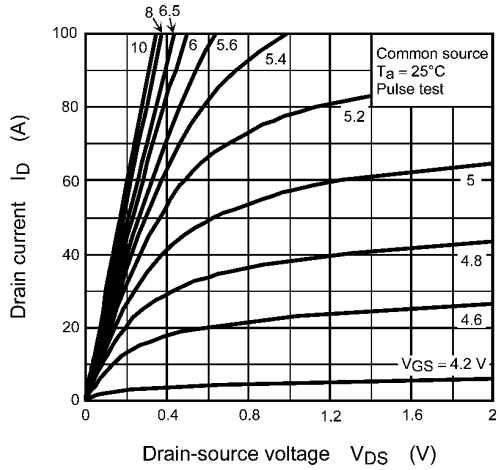


Fig. 8.2 I_D - V_{DS}

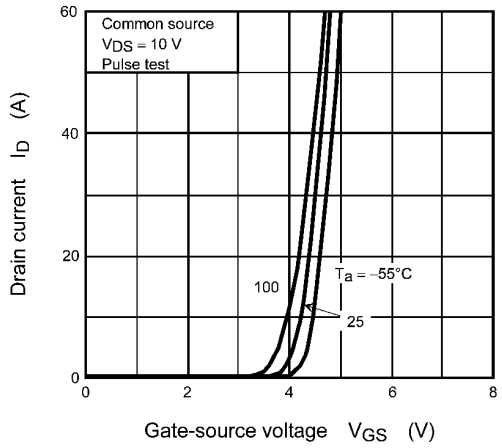


Fig. 8.3 I_D - V_{GS}

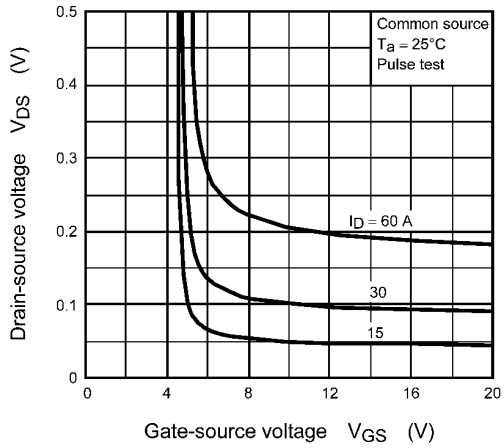


Fig. 8.4 V_{DS} - V_{GS}

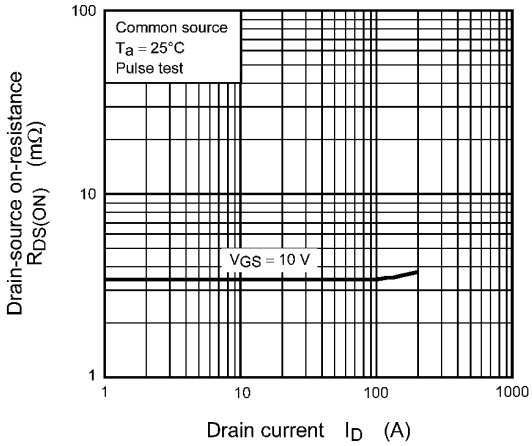


Fig. 8.5 $R_{DS(ON)}$ - I_D

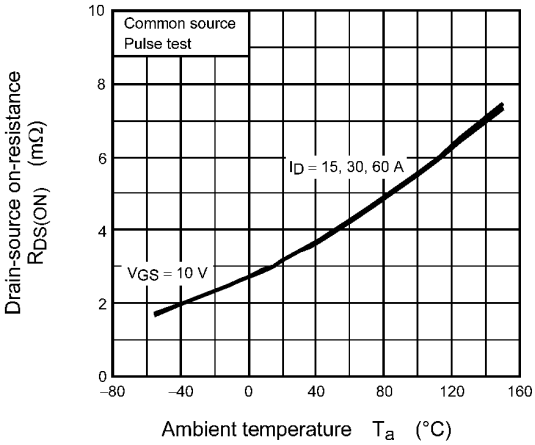


Fig. 8.6 $R_{DS(ON)}$ - T_a

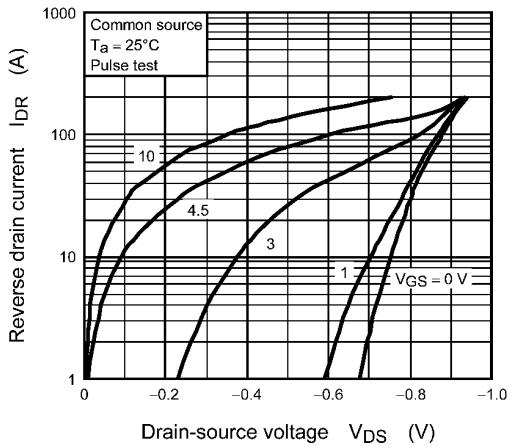


Fig. 8.7 I_{DR} - V_{DS}

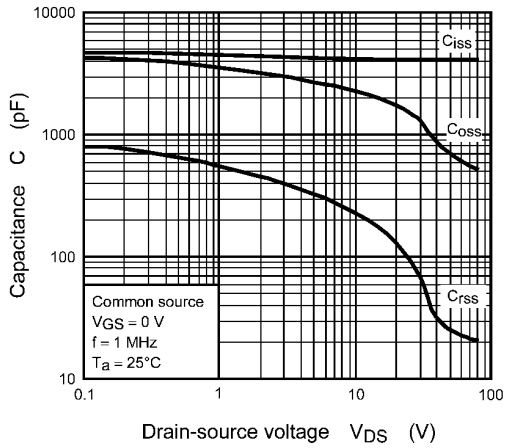


Fig. 8.8 Capacitance - V_{DS}

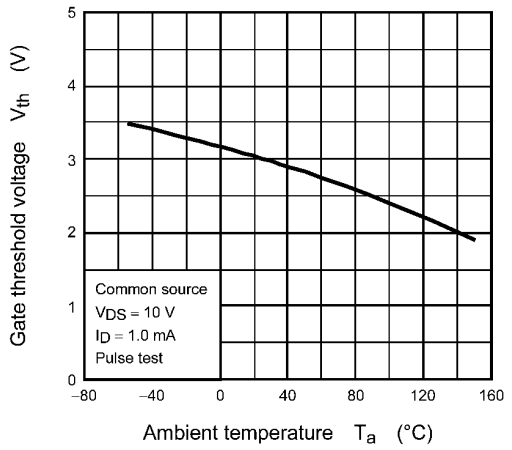


Fig. 8.9 V_{th} - T_a

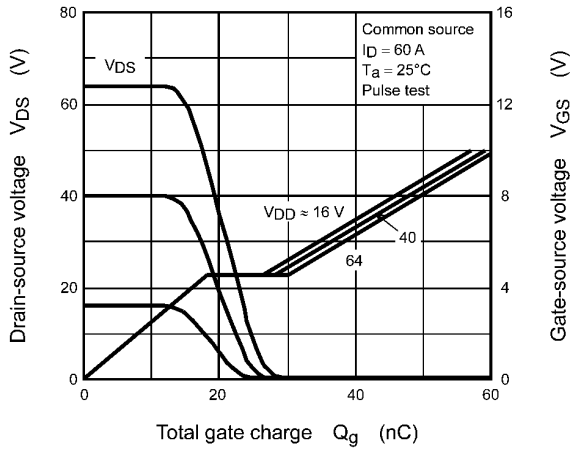


Fig. 8.10 Dynamic Input/Output Characteristics

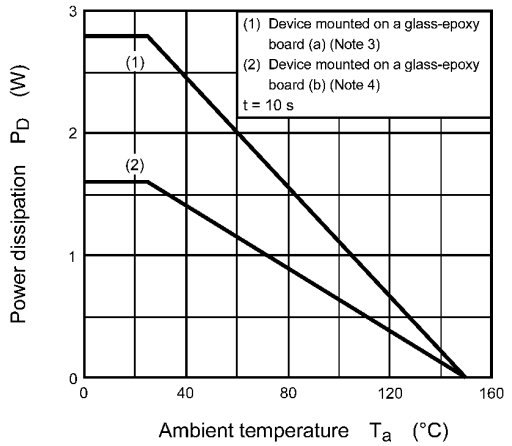


Fig. 8.11 P_D - T_a
(Guaranteed Maximum)

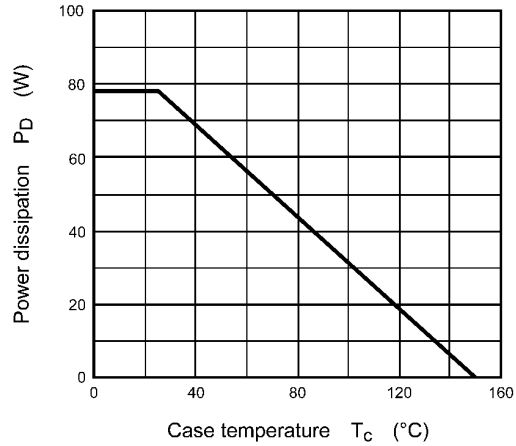


Fig. 8.12 P_D - T_c
(Guaranteed Maximum)

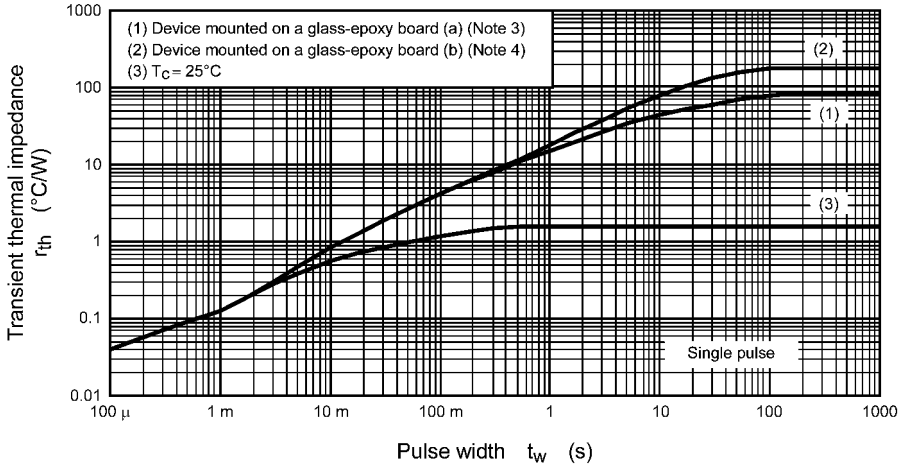


Fig. 8.13 r_{th} - t_w (Guaranteed Maximum)

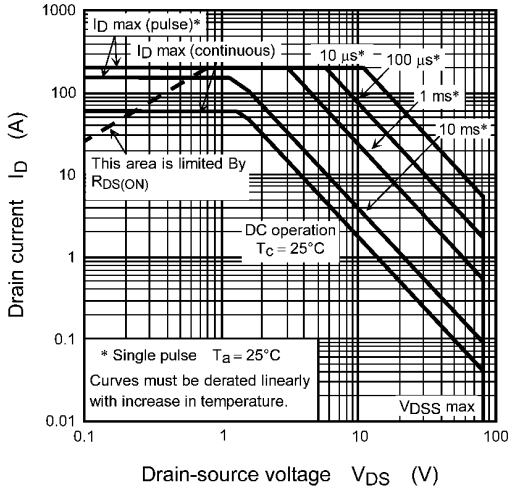
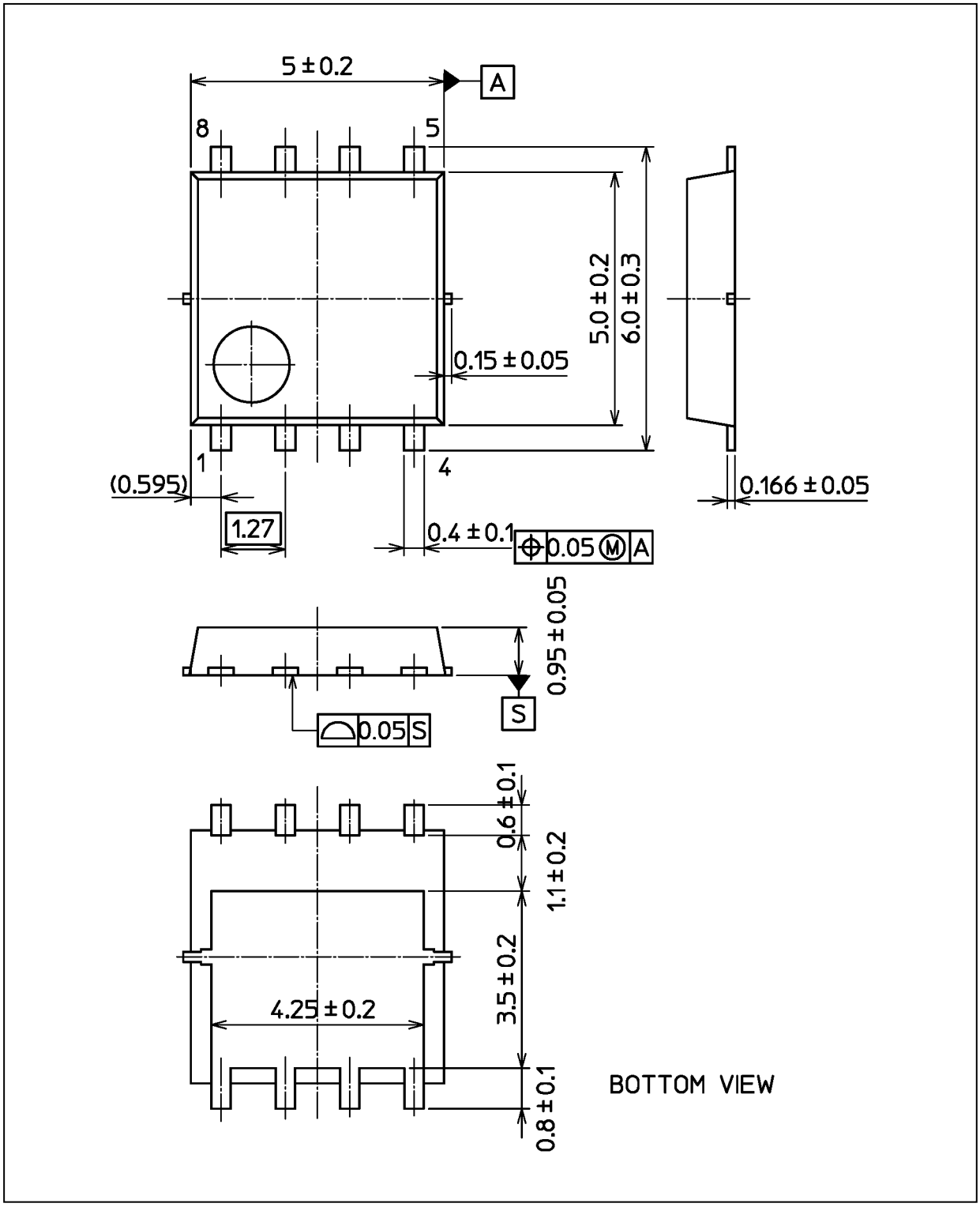


Fig. 8.14 Safe Operating Area (Guaranteed Maximum)

Note: The above characteristics curves are presented for reference only and not guaranteed by production test, unless otherwise noted.

Package Dimensions

Unit: mm



Weight: 0.069 g (typ.)

Package Name(s)
TOSHIBA: 2-5Q1S
Nickname: SOP Advance

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